QUESTION 14: FREQUENCY DISTRIBUTION OF CUSTOMER AGES

14. Scenario: You are a data analyst working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a Pandas data frame.

Question: Develop a code in python to find the frequency distribution of the ages of the customers who have made a purchase in the past month.

Aim: To calculate the frequency distribution of customer ages using Pandas.

Algorithm:

1. Import Pandas.
2. Create a DataFrame containing customer ages.
3. Use value_counts() to calculate age frequencies.
4. Display the results.

Code:

```
import pandas as pd

data = pd.DataFrame({

   'Age': [20, 25, 20, 30, 25, 20, 35, 30]

})

frequency = data['Age'].value_counts().sort_index()

print("Age Frequency:")

print(frequency)
```

Input:

```
Ages = [20, 25, 20, 30, 25, 20, 35, 30]
```

Output:

Age Frequency:

20 3

25 2

30 2

35 1

```
>>>
========== RESTART: C:/Users/sravy/Downloads/DSA03-FODS/ex13.py ==========
Age Frequency:
Age
20    3
25    2
30    2
35    1
Name: count, dtype: int64
>>>
```

Result: The frequency distribution of customer ages was calculated successfully.